Software Engineering — Lab 6

Code and Branch Coverage Analysis

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Section: E · PES University, RR Campus

Repository: L6_L7_08_ski_resort_pass (`ski_resort_pass.py`)

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1. Objective

Measure and improve the test coverage of `ski_resort_pass.py` with `coverage` and `pytest`: first to at least 90% line (code) coverage, then to at least 90% branch coverage.

2. Application Under Test

- **`process_resort_bookings(groups, facility_fee)`** prices each active group by pass duration (season 300, full day 80, half day 50). Children under 12 pay 60%, seniors 65+ pay 70%, blackout overrides pay 50%, gear rental adds 35 and a lodge locker adds the facility fee. More than 4 rejections returns `AUDIT_REVIEW`, zero revenue returns `ZERO_SALES`, otherwise `COMPLETED`.
- **`check_chairlift_anemometer(wind_streams)`** returns `SUSPEND_LIFT` if any lift records two consecutive readings above 45 mph, `NO_DATA` for empty input, otherwise `SAFE_OPERATION`.

3. Commands

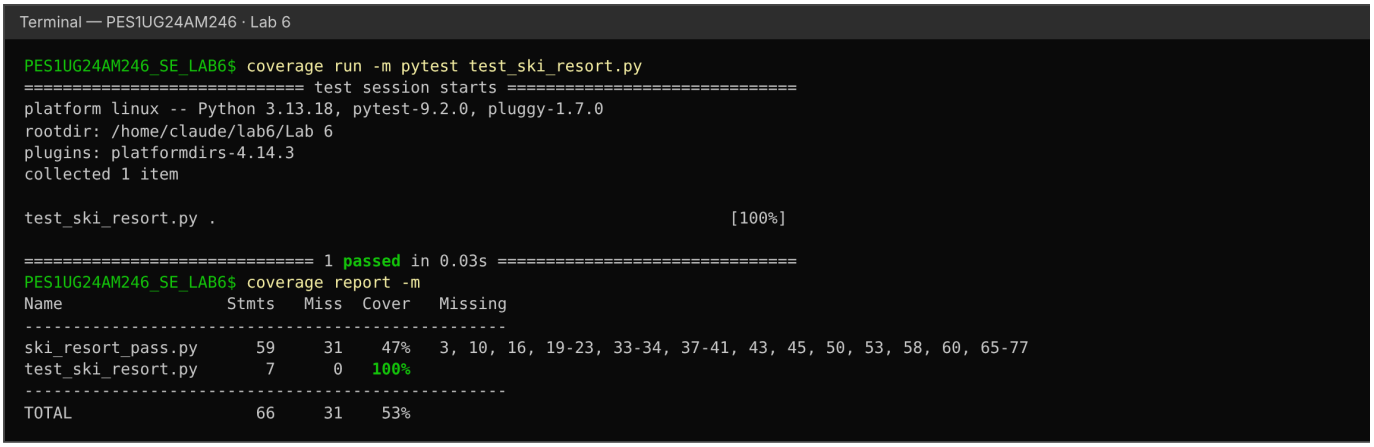
```
pip install coverage pytest
coverage run -m pytest <test_file>.py            # line coverage
coverage report -m
coverage run --branch -m pytest <test_file>.py   # branch coverage
coverage report -m
```

4. Results Summary

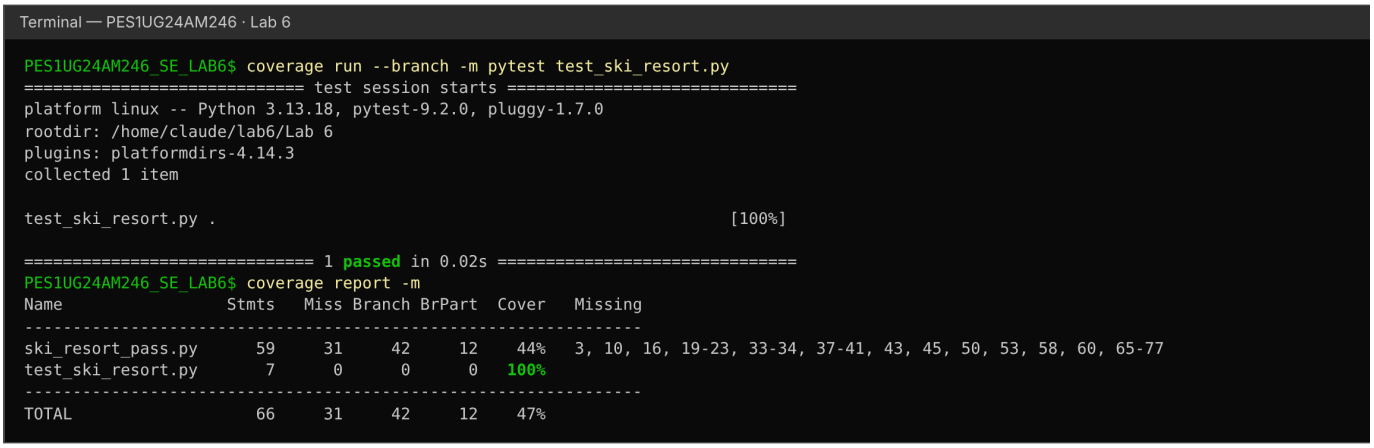
Test file	Tests	Line coverage	Branch coverage
`test_ski_resort.py` (starter)	1	47%	44%
`CodeCoverageTest_PES1UG24AM246_ski_resort_pass.py`	8	**100%**	99%
`BranchCoverageTest_PES1UG24AM246_ski_resort_pass.py`	46	**100%**	**100%** (42/42)

5. Initial Coverage (starter test)

The single starter test prices one adult on a full-day pass, which reaches only 47% of statements. Input validation, the other durations, rejections, blackout handling, the age discounts, gear, locker, the non-`COMPLETED` statuses and the entire anemometer function were never executed.



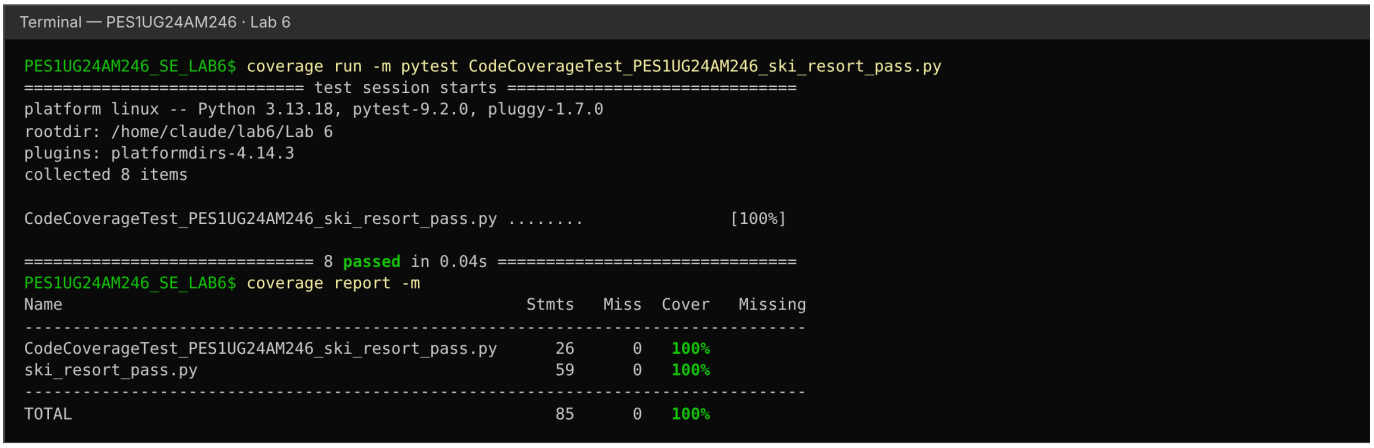
Initial line coverage



Initial branch coverage

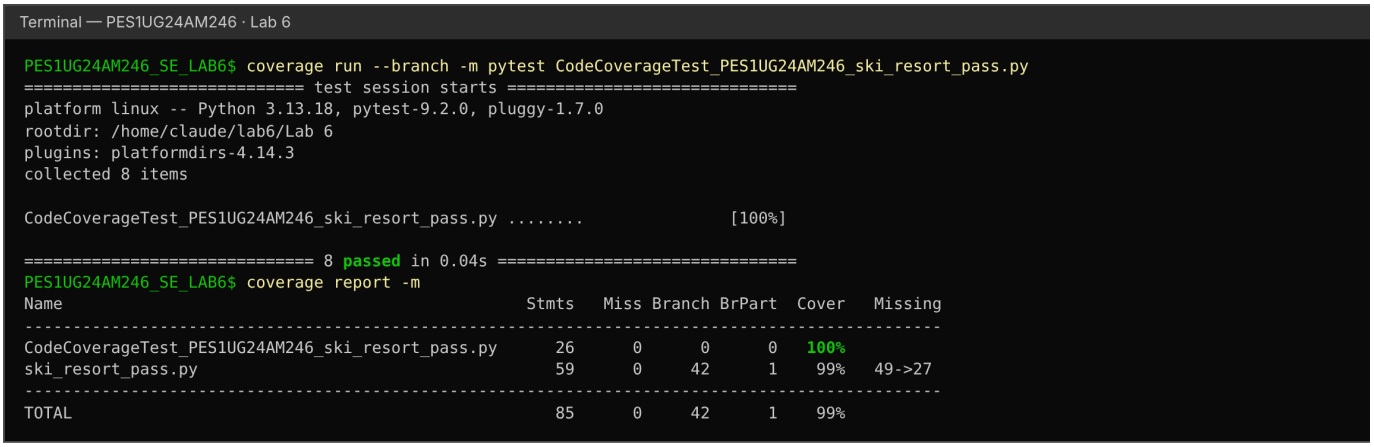
6. Code (Line) Coverage — 100%

`CodeCoverageTest_PES1UG24AM246_ski_resort_pass.py` adds 8 tests. These run invalid inputs, one combined booking that executes every pricing line (inactive group, season group with a child, senior, adult, invalid age and blackout skier, a half-day blackout, an unknown duration and a full-day group), the audit and zero-sales statuses, and three anemometer cases.



Code coverage — line

Line coverage reaches 100%, but measuring branches on the same tests shows **99%** with one missing branch, **`49->27`**. Every skier in these tests rents gear, so the False outcome of `if gear_rental:` is never taken. This is exactly the gap the handout describes: every line runs, but not every decision outcome is tested.

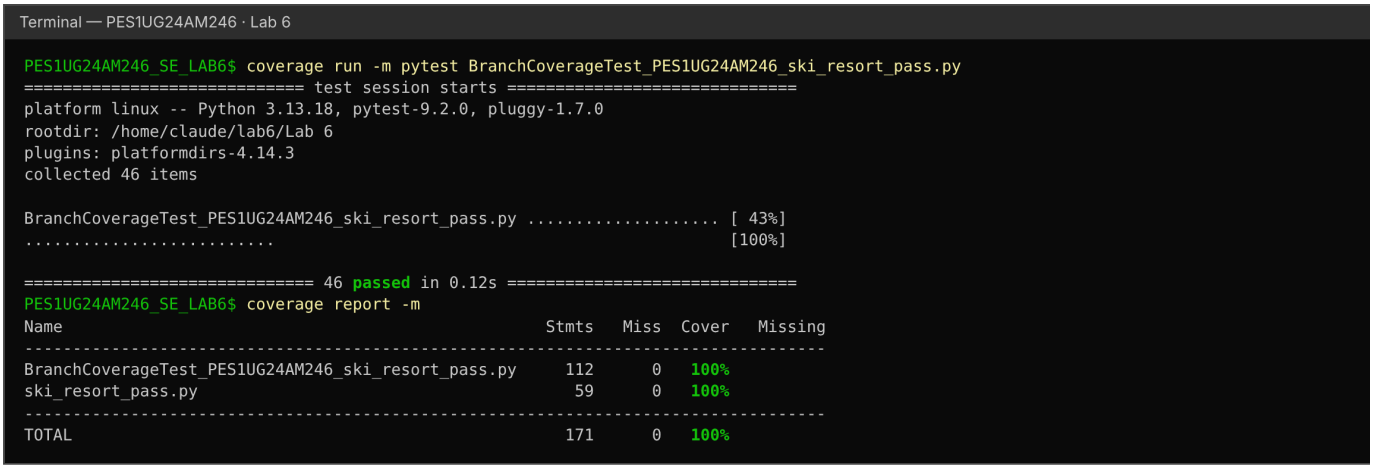


Code coverage — branch

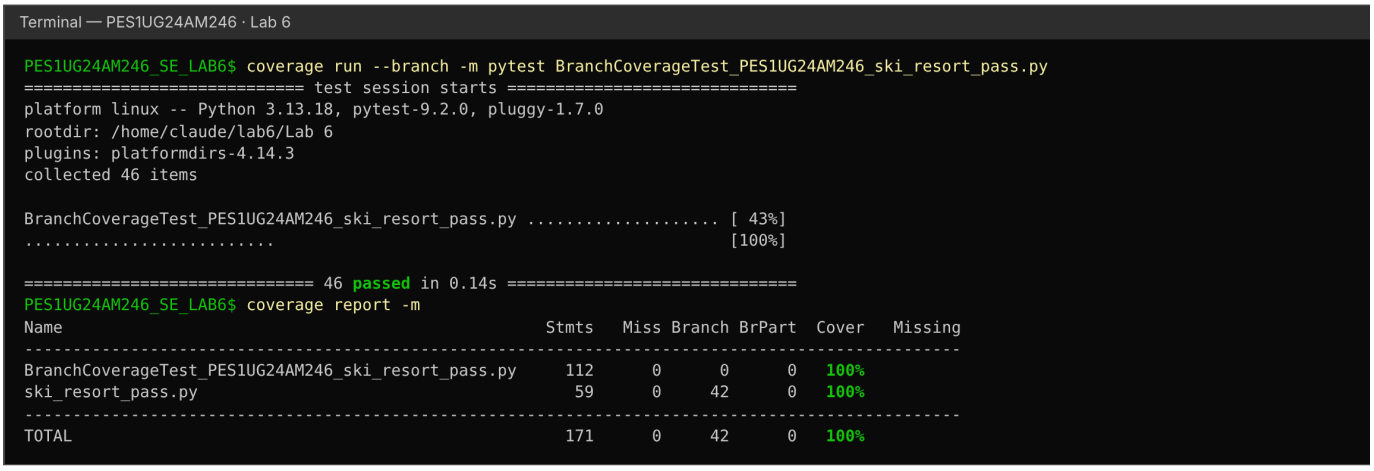
7. Branch Coverage — 100%

`BranchCoverageTest_PES1UG24AM246_ski_resort_pass.py` (46 tests) drives every decision both ways and adds boundary values:

Decision	True case	False / boundary cases
`not isinstance(groups, list) or facility_fee < 0`	string input; fee −1	empty list; fee exactly 0
`for group in groups`	normal groups	empty `groups` list
`not group.get("is_active")`	inactive / missing key	active group
Duration chain	season, full_day, half_day	unknown `"weekly"`; missing key defaults to full_day
`for skier in skiers`	normal skiers	empty / missing `skiers` (locker only)
`age <= 0`	age 0, −5, missing	age 1
`blackout_override` → `half_day`	half-day reset + break	full-day and season half price
`age < 12` / `age >= 65`	ages 1, 11 / 65, 90	ages 12, 64 (adult)
`gear_rental`	+35	**no gear (closes `49->27`)**
`includes_lodge_locker`	+fee	no locker
`rejected_passes > 4`	5 rejections, across groups	exactly 4
`total_revenue == 0.0`	zero sales	revenue > 0
`speed_mph > 45`, streak `>= 2`	46, 46 suspends	exactly 45; single gust; reset; streak not carried across lifts; empty readings



Branch coverage — line



Branch coverage — branch

8. Observations

- **Line coverage overstates test quality.** 100% line coverage still left a decision outcome untested; only branch coverage exposed it.
- **Suspicious behaviour found while testing:** a half-day blackout skier sets `group_total = 0.0` and breaks, which wipes out what earlier skiers in the group were already charged and skips later skiers entirely. The locker fee is still charged afterwards. `test_blackout_half_day_resets_group_and_breaks` documents the current behaviour; it is likely a business-logic bug worth raising.
- **Rejection counting is global,** not per group, so several small groups with one bad entry each can together trigger `AUDIT_REVIEW`.
- **Boundaries matter:** 100% branch coverage doesn't guarantee correct comparisons (< vs <=), so the boundary tests (11/12, 64/65, 45 mph, 4/5 rejections) were added deliberately.

9. Conclusion

Coverage went from 47% line / 44% branch with the starter test to 100% line and 100% branch (42/42 branches) with 46 passing tests. Measuring both metrics showed why branch coverage is the stronger criterion: a suite can execute every line and still miss decision outcomes.